

ActInf Textbook Group ~ Cohort 3 ~ Meeting 7

(Chapter 3, part 2)

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all right greetings thanks for joining everyone it's March 8 2023 meeting seven for cohort three were in our second discussion on chapter three in the textbook and last time we went over the whole chapter kind of ran through it but many things that we can return to and many questions that people have on chapter three so to begin with does anyone whether they wrote the question previously or not do they want to just like share something from their reading about three or some Insight or question about three or whatever else you want to share right now Jonathan and then anyone else yeah I think I'm particularly interested in the the question there about um about the graphic with the distributions um I don't know if uh Francois around at the moment but I was uh talking with him about this this question there are some some things we're a bit confused about so that would be very interesting for me to go through this one the pink one yeah all right thank you any other just general comments otherwise we can just start with this 3.2 all right all right so figure 3.2 so for the top graphs these are Point sampled from distributions μ or x given B for different values of B blanket states on the x -axis this is fine for the bottom left graph what is meant by the average internal state the average value of μ for all the points in the top right graph which are just sampled from conditional distributions so to get exact value we would have to take infinite samples or integrate my question is how exactly do we from the bottom left plot identify the average X and variance around it and then for the bottom right plot what exactly is the expected value that X is conditioned upon the subscript is p of μ given B do we select a specific b or double integrate over B as well great question does anyone want to give a first thought on how anyone can just describe the the graph as they see it or give a thought on how from the bottom left graph the bottom right is generated Jonathan yes maybe I can talk through what what I think and what what the confusion for me was um and so if we go to the the top graphs there um we're we're clearly looking there so X here is sort of for instance some external temperature and B is the the sensory State and so it says um how is the um external

temperature conditioned on the the centuries date even though there's not a causal relationship in that direction and then how is the internal State um uh determined by the the sensory by by the centuries Day by the blanket um and so I think the these two make make sense um and then indeed the the bottom left I think also um makes some sense um in that you can take some particular b um and you can say what are the what's the distribution in both μ and X over B and that will give you this this graph or any particular value of B I think for me the confusion here lies in so the bottom right plot there is written as the expectation um where your where the the distribution is over um it's μ given b um and so it's not quite clear there what um μ given B would normally I would normally think of that as you give it a particular value of B and from that and it's an expectation it's a distribution over the Muse um and the expectation thereby over the Muse but it's not clear here that this is actually for a given B so the question I think is is that expectation at the bottom right really over μ given b or is it just over μ yeah great question I'll try to give a first thought but if anyone sees any other mathematical way go for it I believe it is for a specific B so we're looking at this crosshairs where uh so we'll just continue with this temperature but we're dealing with a with an agent who who's mapping is might be like a little bit wacky with temperature so when μ is 50 that is corresponding to a blanket reading of negative five so if if all we got was a b um reading of negative five our maximum likelihood estimate for μ would be 50. so that's not what we're conditioning on a particular observation of B now conditioning on μ being 50 we have a mean and a variance of just this thank you um Amy I am recording just on OBS not zoom so it didn't make the announcement but indeed it will be uploaded here um so condition on μ being 50. this is where we look at the mean and the variance of this X on μ regression so for μ being 50 X is centered around negative six with a variance that's just literally the mean and the variance of this scatter plot that's the empirical way to say it would just be you just look at the scatter points that are in this column and this is the gaussian that's parameterized from this slice so I I think so that that certainly makes sense I think the question there was so the the μ equals 50 um it's it's saying um I think it says in the in the caption that it's sort of over the over the expectation of the of the music we look at the top right plots that μ equals 50. it's not quite clear where that is coming from I agree I think having another crosshairs here to reflect that that was the B that that gave us the 50 μ the crosshairs being here um it actually uh it does identify across hairs here and it identifies a and then if B is negative uh five let's just say it identifies back onto this line a crosshairs right here which is at surprise surprise negative six plus or minus one okay so really it's sort of flowing around from top left it's

saying the average temperature outside is -6 or thereabouts what does that correspond to in terms of B what does that correspond to in terms of μ and then you get the distribution from that yes perfect and that's how the market blanket even though it's defined as no uh no direct influence of internal and external States no tele no telepathy no telekinesis so all perception action mediated through the blanket yet we still are able to talk about internal States as reflecting beliefs about external States but with this partition and all of the partitioned States internal external and blanket in the simpler plural partition internal external with the Friston blanket with perception and cognition as the two kind I mean sorry perception action as the two kinds of blanket States so here's just a sensory example awesome yeah any other thoughts or question on this one definitely helpful to have that crosshairs uh Amy or Ali uh well actually I think I kind of disagree with with the final conclusion that uh there need to be a Crosshair in the top um two diagrams because uh I don't know I I might have um misunderstood this but I think uh the top two diagrams actually shows uh you see here uh one axis um it's just a blanket State and the other axis is external and in the other one is just the internal stage so here I think the blanket state is generated from averaging over a generating function that generates that blanket States so but in the bottom left diagram we deal with some specific States I mean so for so for instance um here uh what exactly is the value of x when the internal state is say 50 right but I don't know if uh it makes sense to add those crosshairs uh into the final into the top two diagrams given that they just show the the statistical shape of the blank Estates with respect to external States and the internal States I'm not sure where we uh diverge in our conclusions but yeah that I mean this figure as it is uh somehow makes more sense to me than uh if there's a there's some added crosshairs in the top two diagrams I think I hear you on this these are learned conditional associations on the top and this is the per this is the actual particular inference question so it doesn't necessarily clarify it but it does kind of locate us in point space with x b and μ since these are like different views on the same particular partitioned system yeah I think the the reason I think it's important is because when it says here e um the expectation value taken over the distribution of μ given B this is actually for a particular value of B that has been chosen and from the bottom left plot you can't see what that b is it's actually for a specific value of B if I understand correctly yeah it is definitely for for a specific value so cool good question otherwise this would have a different shape it wouldn't just be a gaussian around this it would be like kind of like a mountain range okay just gonna on page 52 the sum part gave me the impression that perception is actually a passive process it brings action to the game but it is not active itself doesn't Maria do you want

to add more or we'll look at 52. it's actually the impression I had when I was reading it so I'm not sure if I if I had the wrong impression or just the the right thing from Thomas park at all is different from the writing from Jacob Howie for instance because Jacob Howard would say that perception and action is more or less the same thing so I don't know yeah this is a great and and uh subtle point so we're talking about um generativity and action and perception are these the same or are there distinguishments amongst them so it's an interoceptive sensor with glucose or neuropeptide or something and then we're doing inference on our hunger status so that's um so for any given beliefs about our hunger and for any given Precision or any given interoceptive input we're going to get like some surprise value so surprise value is just how surprising that data point is given the priors it's in some information theoretic unit so whether you're as minimally surprised as you can be or whether it's super surprising there's some surprise value associated with that observation that um that discrepancy that surprise value can be seen as the um the compromise between like this actively generated top-down prior and and a received sensory information so there's sort of a generativity there's the tale of two densities purely perceptually but we're not talking about motion or movement yet so perception can describe surprise and you can learn you could update you say oh maybe I am hungry or maybe I'm not hungry so you can modify your top down prior which is like a form of covert action to reduce surprise however it is not inactive action so a purely perceptual like a predictive processing model doesn't have to have action like in live stream 43 um that Maria and I did most of the paper was on the predictive processing in the purely inferential case so you're still a top-down priors bottom-up information and then just in like the last section it's like okay now we're going to throw in action as a parameter it's active inference so I think that this is saying in the purely perceptual predictive processing setting it is still the case that surprise is being like um just conversationally actively generated but that active generativity or responsivity of the predictive processing hierarchy is uh not necessarily exactly the same as going out to get dessert and inactive inference there's like the learning and perception way to reduce free energy and then there's the actual action selection part now there might be special cases where the only actions that the agent has are internal or covert affordances like attention and learning but in the general case and in the most um kind of manifest form of action usually we're talking about bodily action does that address it Maria yes I think that the main difference is predictive processing and activities that was microfusion yeah because I had the impression that perceptual perception was almost like a computer that was Computing the KL Divergence between beliefs and Sensations whereas the action was another thing but you

know again I think that I was misunderstanding the differences between predictive processing and active inference yeah I mean they are um active inference is a predictive coding or predictive processing architecture where manifest action or internal cover is itself a parameter that we do that kind of inference on but you could also have a purely passive predictive processing system like no isocades but still talking about priors and perception on vision and then one could even argue that that model still has like kind of this generativity because there is the top-down generated visual um what is perceived but that's not the same type of action or the same level of embedding with bodily action as if your predictive processing model of vision included isocating and so that's one of the important uh developments of active inference Beyond predictive processing and Bayesian brain that's the pragmatic turn is like yeah it's not just that there's this generative model that's always learning and updating it's actually one of the features one of the parameters in that generative model has to be self in World consequences of in Worlds from a sensory perspective because the main question that a predictive processing model for example a vision has to address um if you're like in the machine learning it's like well it has to classify digits it's like okay yes it does but the bigger question is where to look and that's the kind of model that biological organisms are engaging with where should I isocade and then yeah we reduce our uncertainty about the visual scene with isicating and then we could classify the digits but if we only set up the task as being like digit classification then it doesn't need to have this active sampling feature um Ali uh yeah I think uh slightly more nuanced um distinction between uh predictive the difference between the predictive processing predictive coding and other Frameworks with active inference uh is laid out uh in chapter 10 especially the beginning uh section 10.4 page 198 but uh just very briefly although in some predictive processing Frameworks such as embodied or action-based predictive processing perception and action can somehow be correlated with each other um if not exactly synonyms to each one but in active inference uh there's um this subtle distinction that says perception both perception and action are inferential processes so they are just analogous in terms of being this kind of inferential process so they're not exactly equivalent to each other as some predictive processing Advocates may claim great yes they're analogous in their computational handling but they're not identical they're not functionally or semantically identical action and perception are difference Concepts but we have a unified imperative that that talks about perception slash learning and action as both being in the common inferential game of free energy minimization which bounds surprise of the generative model um okay a short question does agent need awareness of preferences can preferences be learned I think

this may have come up at a uh um office hours so does the agent need like experiential awareness of preference you can construct models that have this but it's not a feature of the kernel of the model so single layer active model no you don't need to even go there but you could make a nested model where the higher layer is like attention or meta awareness can preferences be learned yes here's a paper of Sajid at all preference learning in general though is a very challenging problem because what if you um how do you even know what you would want to learn so it also quickly moves into this kind of like hierarchical modeling setting where when you write out the model you'll have a lot of clarity about what is being learned and what isn't being learned but in the general um kind of pre-model specification space these questions as with many many others it's like in principle yeah you could whether is or isn't that is going to be about any specific model but it's a great question okay other upvoted ones and then we'll continue down is there a minimal example of a thing defined by a Markov blanket for example a cell seems to satisfy such definition same with proteins and other structures is it correct to say a thing is just the temporary Persistence of a Markov blanket are all things then just non-equilibrium steady state processes as per figure 3.3 do we need to be more strict in saying these things are only self-evidencing things rather than things in the colloquial sense who has any thought on this I'm sure there's a many takes slash ways to say it but the only piece I would add here a thing is the modeled or measured Persistence of a Markov blanket because Markov blankets exist on maps Markov blankets are features of Base graphs there are common features of Base graphs every base graph where it's not fully connected so anytime there are nodes that aren't directly connected there are Markov blankets that insulate those two sets of nodes so it's actually like not a rare or exciting uh event on base graphs and also it doesn't necessarily demarcate the spatial temporal boundaries of an embodied agent the thing as we're going to be talking about it is for the given spatial and temporal scale and signal to noise and all of that that the Observer is using if it's identifiable as a thing even a process thing like a river or something to The Observer then yes it had to have persisted as that thing within that spatial and temporal scale that the modeler was making that map of maps are not Observer independent Mark all blankets are on maps yes every level of organization end of a multi-scale complex systems will fit that definition are all things non-equilibrium steady state processes these things are only self-evidencing brings us to our favorite image in the um strange kinds article so there are things or particles here particle is the same as thing that are inert and so no appeal to self-evidencing is needed to describe their persistence um or yeah I'll say that um there are octave particles whose actions can be understood as self-evidencing again without an

appeal to awareness or a Consciousness or anything like that and this is what is called sentience cueing the ongoing debate again so not qualia sentience but sensing and adaptively acting such as um some uh like the wood lice that Carl identified in his childhood where they were walking faster in the Sun and slower in the shades so they spent more time in the shade and then self-evidencing in the case of these sophisticated or strange particles the self-evidencing includes like self in world with others like me under nested counterfactuals so that's not present in the base of the but the base models that we are going to be looking at especially in like the upcoming chapter this is like the Lego block that these much much more sophisticated models are built upon Scott if for some reason I want to model a rock using this framework I would probably model it as an inert particle yeah absolutely you could even just draw out the particular partition here's the photons coming off of The Rock and they're determined by this other action impinging on the Rock and here's my inference on The Rock's weight or here's the rock on the scale whatever situation you're going to be talking about it um yes it's partitioned from its environment and it's not taking action selection or one could see its action selection choice to stay in one location as like the simplest case of Bayesian mechanics just like a rock physically being in one location is the simplest location thing of uh of simpler than something falling a belief just existing in one location is simpler than it moving Ali yeah uh yeah I think we should probably point out uh the notion of uh blanket index here as well because you see Marco blanket is not something predefined or even uh an established value because we can Define markup blankets depending on how complex the system is and by complex here I mean uh it's uh more or less precise mathematical definition I mean so for as the system becomes more complex and in a more in higher dimensional systems uh or in other words when uh the system has a sparse coupling with its environment the markup the blanket index approaches zero but in less complex system or in other words in the systems that uh doesn't necessarily have this kind of non-linearity embedded in them and they behave more as linear systems uh the mark the blanket index and maximizes so yeah it's uh and it's it's a kind of a spectrum and we can have different levels of uh weakness or um blanket index uh embedded to the definition of our markup blanket yes thank you yeah this is recent uh some advancements and uh but there's a lot of other like discussion of well what about when it's shifting materiality of the boundary or what kinds of and it just there's a lot more to say but it's always good to remember the markup blanket it's not like internal external blanket states are pre-assigned about the world their base graph partitions so the partitions on a given constructed map never the territory it's just a total um apples and oranges apples and maps of

Apple's question mark out blankets just like simply are not in the world that's a good first angle and then we can talk more about like which thing is the person modeling and what kind of particle are they modeling it as and so on but just helps to like instantly jump a two feet into the modeling question which is where you are when you're talking about the Markov blankets um rather than in the underlying ontological what is the the true substance of what the thing is which is actually still a question that many people have found that the Marcon helps them get around not to bypass but like to to be sense making in that space so it's not even that it isn't useful for that space it's just that the markout blanket is something about base graphs not about whatever territory you're trying to describe with these graphs um living organisms resolve this fundamental biological Problems by exerting active control over their states which range from every biological mechanism in the world is a way to understand why automatic mechanisms are active in thinking that these still involve calculating efe over future States but the time Horizon of automatic reflexive behaviors is small is there another concept of passive control or is all control active so and then there's an example provided involving like a the so-called reflex arc so a few answers here first in chapter five we will be looking forward to that essentially exact example where depending on the dopaminergic Precision and tone in this dopaminergic region of the mammalian brain there's going to be two Paths of policy selection there's going to be one path of policy selection that basically passes habits forward and there's going to be another path of policy selection where expected free energy G is used to sharpen action priors habits into action or policy posteriors which are updated um sharpened according to the extent that they serve to minimize free energy so that's a dopaminergic mediated handoff between like what might be seen as sort of just habitual passing forward what has been done versus taking a different lens according to a defined G functional and then to the question of the kettle here's a predictive processing architecture by which um the spinal cord this is like a cross section of a mammal spinal cord with a butterfly shape incoming sensory data can be reflexively acted upon without getting passed into this entire cortical hierarchy one key piece to remember just like Markov blankets are about maps not territories generative models calculate expected free energy not organisms is there a way to think about it that it involves calculating efe yeah if you write the generative model so that um blood pressure regulation involves an efe calculation the answer is yes if you write the generative model so that blood glucose or blood pressure regulation does not involve efe the answer is no so in principle anything's possible but when it really comes down to it it's always going to be about how you specified your generative model or your thought experiment anyone can also give a thought on

this but some um might say like control theory is the application of action that's the line between like signals and control theory if you uh you really like the temperature to be uh 25c and you have a Time series model or variational Auto encoder of temperature all these different things and then sometimes you're like oh it's 25 it's perfect other times it's too hot or too cold like so you're in the single processing world and you still might have a preference about what signals you're coming um to infer but this kind of comes to Maria's question from earlier there's a and so that can still be a predictive processing architecture but when we're talking about control even though this term often has connotations of like um commandeering or like excessive control that's just a cultural connotation it's really the distinction between a signal processing perceptual only model which still can have that Tale of Two densities and then what is called control theory which is basically any introduction of Behavior and that introduces like well why are we acting well two in this framework not maximize reward According to some proposed reward function but to reduce the Divergence between our sensory observations and our preferred sensory observations foreign model evidence $p(y)$ is mathematically maximizing model evidence P of Y so we have a portfolio we have a given data set why and we're going to have a portfolio of models each model can then be said to have a certain amount of evidence in the sense like it corresponds to could generate that data better or it recognizes that data with less variance and we want to select in this situation the model from our portfolio or like families of models that we swept over we want to select the models that have the highest model evidence and that is equivalent to being to selecting the model that minimizes our surprise about incoming sensory observations so it's like if temperatures being generated a certain way we have certain temperature data set one way to think about this problem is we want to select the mean and the variance for the gaussian that have the most evidence with respect to the data we have another way to think about this exact same situation is we want the gaussian such that surprise is minimized as new observations come in surprise is defined as the negative natural log of model evidence which has like a lot of features and we'll kind of come back again and again to these features of logs and then as to the specific question um it's a sometimes we call it a fancy I but it's called fracture so it's a i for information even though it looks kind of like a j but it's fracture is the name of the fonts family importantly the agent's generative model cannot simply mimic external Dynamics what is meant by this statement does this mean the agent's prior does not need to be updated since it has a perfect model already if so how could an agent Define by their generative model in the markup blanket have a perfect model when what is available is um its local sensory States this

is in the neighborhood of the good regulator theorem from cybernetics so adaptive cybernetic systems have to be able to have like the requisite diversity but also the ability to be good Regulators which means that they have to uh not necessarily make like a digital twin cognitive model of their environment it doesn't need to be uh bit for bit Adam for atom recapitulated however in challenging environments the agent will have to act as if it contains a generative model that includes some of the latent and unfolding features of the niche again the temperature example if our only imperative of the generative model was to fit temperatures accurately as possible you would just fit temperature accurately as possible and then when it left your physiological balance you just die so we wouldn't see that kind of organism around so the agent's generative model has this dual imperative of accurate but not overfit perception and adaptive action that's the synthesis of the signals and information processing incoming side with the action selection control theoretic outgoing side because simply mimicking will not lead to adaptivity it's just not adoptive agents need or want to do but you can contrive an in silico example where it does have a perfect internal model of the external States or it could be a fully observable situation like a game of chess where like the board is fully observable now you can't observe into your opponent's mind but again you could imagine an in silica environment where like there was that kind of visibility so there's other aspects to this but basically the agents generative model doesn't have to be the exact structure of the external world like temperature in the world could be a continuous variable and the agent generative model could just be like too hot too cold or just right so it doesn't need to mimic the structure of the external world but there has to be some semantic resemblance if temperatures indeed something that is relevant for survival there's gonna have to be something like an as if Thermo modeler and again it doesn't need to be lockstep it doesn't mean that it's just recapitulating the internal Dynamics but it has to have a cognitive structure that allows it to have actionable a representation of the external including like the the consequences of different actions because it's not just a purely descriptive model it's also needing to take actions but as we um saw in chapter 2 there's special cases when there's nothing to know or nothing to learn then you have utility oriented policy and if there's no differences in pragmatic utility for different outcomes then you have like novelty or information driven policy so in principle a huge number of special cases Can Be Imagined but in general it's not the case that the GM simply mimics external dynamics that would be a contrived setting is more complex and itinerant Dynamics referring to concepts of multi-stability meta stability criticality yes definitely everything with complexity science and itinerant Dynamics just like continuing to wander maybe never even

revisiting maybe revisiting multiple tractors conditional attractors phase transitions all of these kinds of topics so great question is it correct to say that Gibbs free energy and helmholtz free energy belong to the context of equilibrium stat Thermo and variational free energy belongs to the context of non-equilibrium thermodynamics um for a disciplinary answer ask someone who knows and cares inbox 3.2 what they're highlighting is not necessarily the difference between equilibrium and non-equilibrium though that distinction is here rather I believe that they're highlighting that variational and expected free energy are information theoretic quantities whereas Gibbs free energy describes like how much calories of heat are released when the candle is burnt and so we say oh the candle is burning it's reducing free energy it gives free energy that's what made it an irreversible reaction or spontaneous reaction but the variational and expected free energy are in information geometric spaces they're not referring to the same exact reference of thermodynamic systems like heat bath steam engine and so on it gets blurry because some people are really um pushing like pan cognitivist pan information list it from bit and so people do discuss some relationships at very deep levels between for example information engines and heat engines so it's a deep hole to go into but just from the outside Gibbs free energy is talking about candles burning variational free energy is talking about informational bomb systems Ali yeah uh well again uh going back to uh what we talked in the previous cohort um actually in the last uh couple of years there's this uh push towards a developing a kind of new mathematical formulation of active inference which doesn't require the nest assumption as it's required as its precondition so previously and especially from about 2012 to 2019 most of the active influence literature was focused on developing the density Over States formulation but in the last couple of years the trend has shifted toward developing the density overpath formulation density or past formulation of active inference which it doesn't necessarily require uh NES assumptions so in its most current formulation we can say that active inference is much more General and also includes even the systems that doesn't obey the NASA assumption or requirements yes great and then last question and then we'll look ahead to the coming chapter how does the taking of drugs which flow across the blanket and directly affect the internal States and therefore GM fit within the framework I.E the blanket is semi-permeable it is not only sensory and active so blankets are not in the world you could model and make a model where for example the blood-brain barrier is like uh blanket States and then the question points out right there are actually the materiality of certain substances is such that just because one state was internal with respect to photons in my map doesn't mean it's internal or secluded with respect to some lipophilic drug if you draw out the

base graph like what you're actually talking about in terms of the base graph of what what information flow because the base graph is not identifying the spatial temporal or the lipid bilayers it's not just Anatomy it's actually more like an informational physiology because it's describing connections amongst different parameters in the map not necessarily different like physical articulations in the anatomy however more broadly like this concept of semi-permeability that is being addressed with the blanket indexed and as well as Dynamic blankets and it's not sensory in active states which affect the internal States in reality again in the reality of the particular partition on the base graph this is the definition of what the blankets are however whether the labeled and assigned nodes in the map have this or that relationship to different embodied territories is the map territory question so we can definitely return to these topics but I think by drawing out a base graph there'd be a lot of clarity around like what is internal or external to what it's not like there's like a highlighter that you can just tag objects in the world this is an internal State and that's an external State and that's a blanket awesome yes it's like correlation is not causation there's just a few where on like the the 700th time hearing it's like oh I guess it is let's look ahead to chapter four so we are in chapter three High Road notice that chapter 2 and chapter 3 are the low road and the high road to octave inference so we haven't begun active yet but we've gotten to there but right when we were on the outskirts of the city we like pulled away um I I don't know not not really but kind of chapter four is the generative models of active inference so this is going to be like the heart of active after we've taken it from these two um roads the low road is the how with a Bayesian statistics which doesn't have to be about adaptive Auto poetic anything Bayesian statistics could just be about the temperature and then we talked about it from The High Road perspective this imperative for self-organization and persistence which doesn't have to be implemented with Bayesian statistics these two roads come together in the generative models that we construct in active inference which use the Bayesian statistics and Implement a Bayesian mechanics chapter 4 complements the previous conceptual treatments with a more formal treatment we're going to be talking about free energy and Bayesian inference and specifically the generative models that are used in active inference 4.2 Bayesian inference to free energy um previously there was a broad framing with a Bayesian brain which has been accepted for some decades if anyone has like a doubt or comes across a doubting person for sure ask for an alternative and always be open but it's like as opposed to what the frequentist brain so Bayesian brain is a natural framing not too constraining at all it just brings Neuroscience in line with modern statistical methods and if we take the inactive turn or the we want Bayesian brain plus action which is

basically octave inference um they're going to return to base theorem figure 4.1 we're going to have a bit of a discussion on Jensen's inequality and some properties of logs so for any of the current or X math professors in this group any thoughts you have on logs and Jensen's inequality would be helpful but we're going to come back to it more equations some transformations of logs and model evidence yes exactly the two people who I reference sounds great that log discussion is going to yield the KL Divergence section 4.3 is going to be about generative models so to calculate free energy we need three things a data a family of variational distributions and a generative model well the cell is minimizing free energy it's like does the cell consist of data and a generative model no oh it's lipids and proteins okay it's not minimizing free energy your statistical model of the cell consisting of these three ingredients can minimize free energy so that's like a key map territory piece figure 4.2 is going to be like a step-by-step walkthrough really really helpful to think about this grammar of Bayesian causal graphs this is like one event that causes another event this is like one event that causes two events this is like two things that cause one outcome and this is like a chain of V causing X causing y so you can think about examples in our own life but this is like the grammar of what these nodes and edges mean in base graphs figure 4.3 is a classic figure it displays on the top the discrete time generative model in active inference which is called partially observable because there's this hidden state in and it's called a Markov decision process because there's this markovian time property which means that the past only influences the future through the present and it's a decision process because there's this control theory aspect so that's a partially observable Markov decision process and then on the bottom is a continuous time generative model discrete time is focused in chapter seven continuous time is focused in chapter 8. 4.4 is a precursor to chapter seven it's about octave inference in discrete time more details 4.5 is a precursor to 8. it's a precursor to talking more about continuous time generative models some more details on Marco blankets continuous time models generalized coordinates of motion comes up a lot it's like instead of just representing the location in GPS of the car it's like location and all of its derivatives so its velocity acceleration and so on that turns out to be really important for certain dynamical systems back to imp as predictive coding Plus Motor action we'll unpack some of these figures and summary so chapter 4 gets to the two big families of generative models that are used in active inference so everything we've been talking about with generative the two biggest families like vertebrates and invertebrates or any other taxonomic division you want are basically represented in figure 4.3 and they can be hybridized too which happens in chapter eight so that's chapter four we will um return next

week go through the chapter looking forward to any questions um and last Point Jonathan there's a lot in chapter four have you found two weeks I've been sufficient to get the grips with it yes various chapters like it's it's one reason why we take the cohorts multiple times because it's not just two weeks it's not like if it were just six weeks there'd be enough to get to grips with it necessarily so going through it multiple times and uh just carrying on with the pace we're not grading or testing obviously so just continuing on yeah only two weeks deep ramju um so just carrying on with the schedule and just trying but but if but we can definitely think more about this in terms of uh future cohorts any feedback you ever have you can add it here but many many things could uh happen you know a mathematical pre-learning or a supplement or um a one week per chapter accelerated or a three week per chapter we really don't know so thanks everybody all uploads recording so see you later